

Twenty-seventh day of class - 12/2

Fun fact: The only three rational points on the curve

$$X : x^3y - x^3z - y^4 - 2y^3z + y^2z^2 + yz^3 = 0$$

are $(1 : 0 : 0)$, $(0 : 0 : 1)$, and $(1 : 1 : 0)$.

The curve X has genus 3, and there is a map from $X \rightarrow E$, where $E : y^2 + y = x^3 + 12$. However, this is not helpful because E has positive rank, and it's not easy to know which $P \in E(\mathbb{Q})$ could be the image of a rational point on X .

However, there is also a map from $X \rightarrow E'$, where

$$E' : y^2 = x^3 + (15 + 6\sqrt{-3})x^2 + (24 + 24\sqrt{-3})x.$$

If $K = \mathbb{Q}(\sqrt{-3})$, the group of points $E'(K)$ is finite (it is a cyclic group of order 6). Using this, we can determine all the points on X with coordinates in K (there are 5) and the rational points are a subset of that.

Review worksheet

The problems below may give you a sense of the sort of thing I might ask on the final exam. Another source of problems are old final exams (from 2015 and 2017) which I have posted on Canvas. (I tried to avoid problems from these exams, but didn't quite succeed.)

1.

(a) Are there any rational points on the conic $x^2 - 3y^2 = -1$? If so, parametrize them. Do this computation without any assistance of Magma.

(b) Are there any rational points on the conic $x^2 - 2y^2 = -1$? If so, parametrize them. Do this computation without any assistance of Magma.

2.

(a) Let $E : y^2 = x^3 + x + 1$. Add the points $P = (0, 1)$ and $Q = (1/4, 9/8)$ on E . Do this computation without any assistance of Magma, but you may check your answer when you are done.

(b) Let $E : y^2 = x^3 + x + 4$. Let $P = (0, 2)$ be a point on E . Compute $2P = P + P$. Does the point P have finite order or infinite order? Do this computation without any assistance of Magma.

3. Find rational numbers x , y and z so that $x + y + z = 0$ and $xyz = 346$. You may use Magma to do any computations that you like.

4. If E is an elliptic curve over $\mathbb{C}$, how many points in $E(\mathbb{C})$ are there that have order exactly 4? You may use Magma to check your answer, but the justification you use cannot rely on any computations in Magma.

5. Use the Lutz-Nagell theorem to determine the torsion points on $E : y^2 = x^3 - 4x + 4$. You use Magma to do arithmetic, compute divisor, factor polynomials, and add points on E , but you may not ask for the torsion subgroup or the Mordell-Weil group.

6. Let $n = 10^{100} + 1$ and $E : y^2 + x^3 + nx$. You are given that $E(\mathbb{F}_7) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ and that $E(\mathbb{F}_{17}) \cong \mathbb{Z}/26\mathbb{Z}$. Determine the torsion subgroup of E . Do not use Magma on this question.

7. Your friend told that their favorite elliptic curve has rational points on it with x -coordinates of $-36, -26, -8, 4, 28, 34, 64, 244$ and might have additional points. Can you conclude from this that there are infinitely many rational points on the elliptic curve? Why or why not? You can use Magma on this question, but I do not expect it to be useful.

8. Suppose that Γ is an abelian group together with a height function $h : \Gamma \rightarrow \mathbb{R}$. Suppose that Γ and h have the following properties:

(a) For any real number M , the set $\{x \in \Gamma : h(x) < M\}$ is finite.

(b) For any $P_0 \in \Gamma$, there is a constant κ_0 (which depends only on P_0) so that for any $P \in \Gamma$,

$$h(P + P_0) \leq 2h(P) + \kappa_0$$

(c) There is a constant κ so that for all $P \in \Gamma$, $h(3P) \geq 9h(P) - \kappa$.

(d) The index of $3\Gamma = \{x + x + x : x \in \Gamma\}$ in Γ is finite.

Show that Γ is finitely generated. (I don't anticipate Magma being useful.)

9. Compute the rank of $E : y^2 = x^3 + 13x$. Do not use Magma on this question at all.

10. Let $E : y^2 = x^3 + 10x^2 - 112x$. Use Magma to compute the image of α and the image of $\bar{\alpha}$. Do not directly ask Magma for the rank or the Mordell-Weil group.

11. Let $f(x) = \frac{x^3 + 2x^2 - 2x - 2}{x - 2}$. Are there finitely many or infinitely many rational numbers (x, y) with $(x \neq y)$ and $f(x) = f(y)$?

On this problem you can use Magma to compute maps between curves, add points on elliptic curves, and factor polynomials. However, you cannot ask Magma to do any of the following: compute the rank or generators of the Mordell-Weil group, search for points on curves, compute the torsion subgroup, or use the command `IsLocallySolvable`.

12. Are there integers M, N and e , not all zero, so that

$$N^2 = 28M^4 - 33851e^4?$$

You may use Magma to do anything you like on this problem.